LECTURE-6

- **Molecular-Orbital Theory (MOT)**

TEXT

Essentials of Physical Chemistry

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Molecular Orbital Theory (MOT)

- It was proposed by *Hund* and *Mulliken* in 1932.

All atomic orbitals of the atoms participating in molecule formation get disturbed when the concerned nuclei approach nearer. They all get mixed up to give rise to an equivalent number of new orbitals that belong to the molecule now. These are called the molecular orbitals.

MOT (contd.)

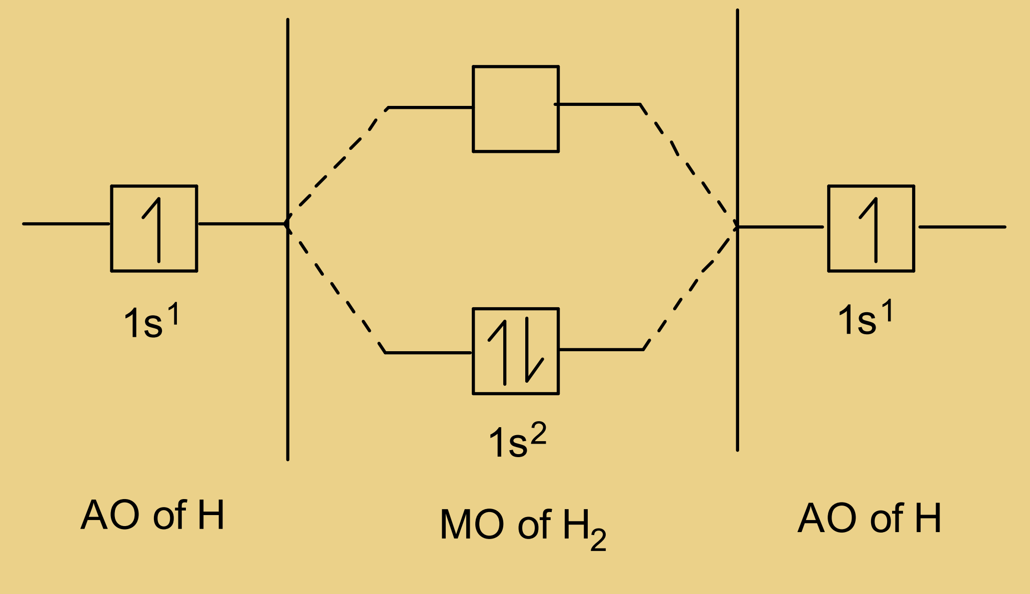
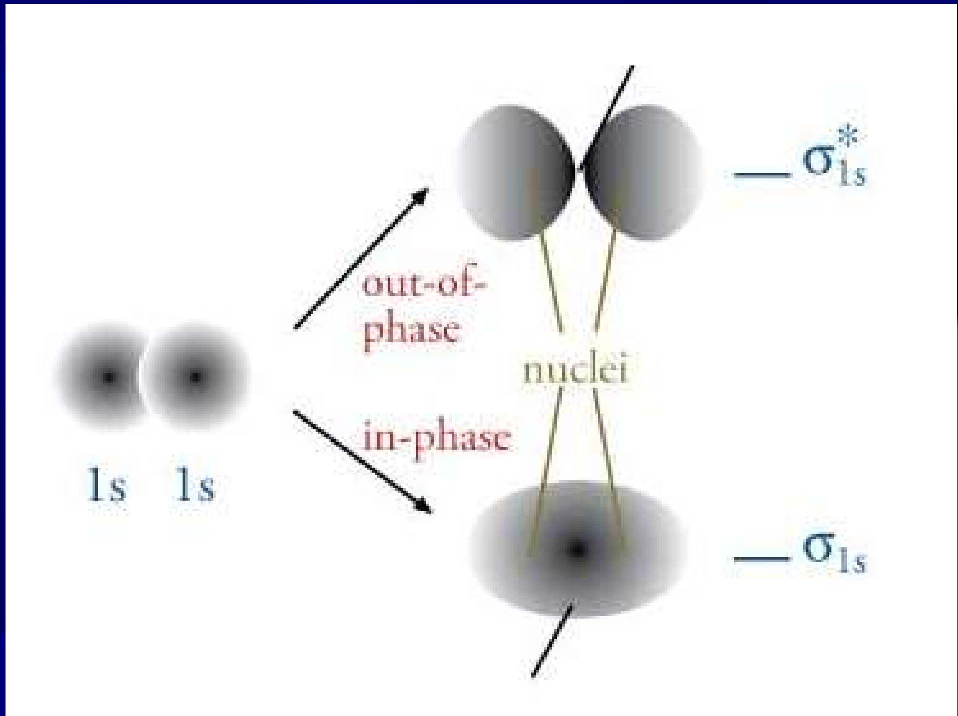


Figure: Formation of H_2 molecule



MOT explain the formation of a covalent bond in a better way.

Main Features of MOT:

- A molecule is quite different from its constituent atoms.
- Atomic orbitals (AO's) of individual atoms combine to form molecular orbitals (MO's) and these MO's are filled up in the same way as AO's are formed.
- The number of MO's is equal to the number of AO's.
- The shapes of MO's formed depend upon the shape of combining atomic orbitals.
- The MO's have definite energy levels.

Bond Order (B.O.)

The number of bonds in a molecule is one-half of the difference of the number of electrons in the bonding molecular orbitals (BMO's) and the number of electrons in the anti-bonding molecular orbitals (ABMO's).

B. O. (contd.)

Mathematically we can write,

$$\text{B. O.} = \frac{(Number\ of\ electrons\ in\ BMO's) - (Number\ of\ electrons\ in\ ABMO's)}{2}$$

$$\therefore\ \text{B. O.} = \boxed{\frac{N_b - N_a}{2}}$$

Information given by BO:

(i) Stability of the molecule or ion:

A molecule is stable if $N_b > N_a$; example- H_2
A molecule is unstable if $N_b \leq N_a$; example- He_2

Information given by BO (contd.):

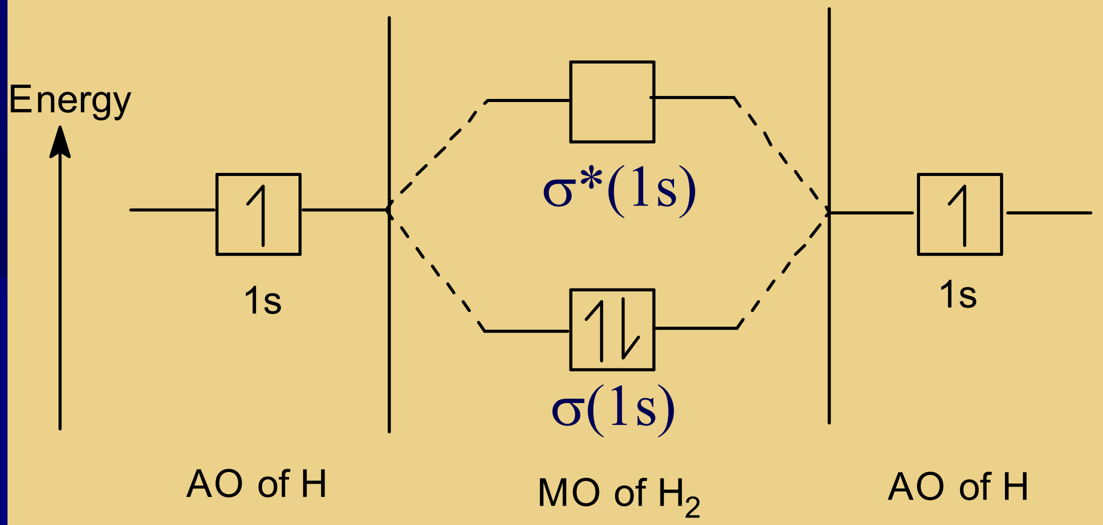


Figure: Molecular orbital diagram- formation of H_2 molecule

Bond order in H_2 molecule is $\frac{2-0}{2}=1$. Thus two atoms of hydrogen are bonded through only one bond in the molecule and hence stable.

Information given by BO (contd.):

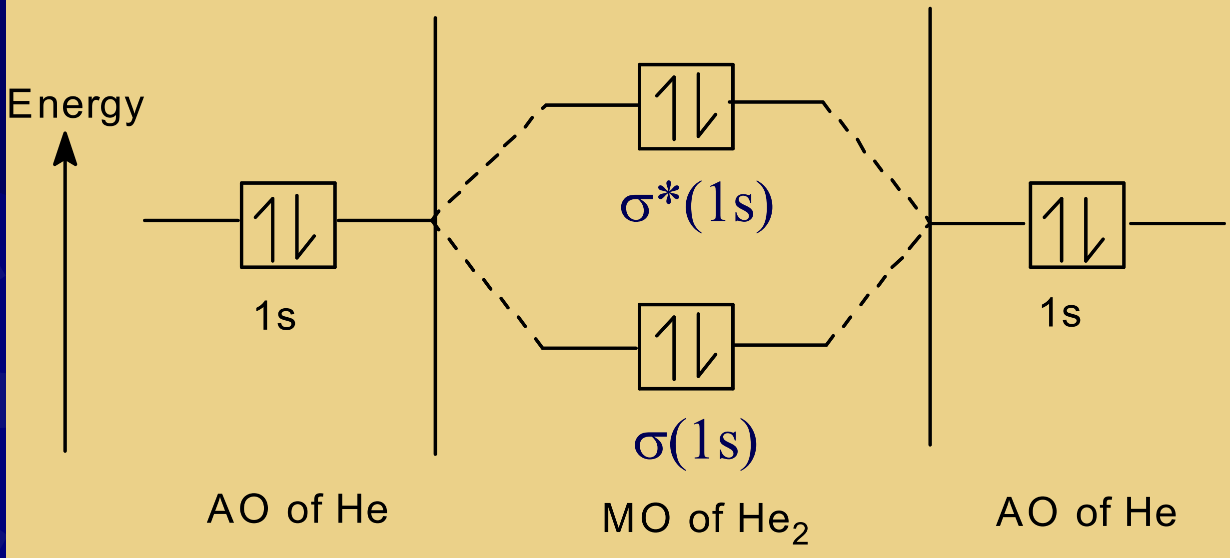


Figure: Molecular orbital diagram- formation of He_2 molecule

Bond order in He_2 molecule is $\frac{2-2}{2}=0$.
Therefore, helium molecule is not formed.
i.e. unstable.

Information given by BO (contd.):

(ii) Bond dissociation energy : Greater the bond order greater is the bond dissociation energy.

(iii) Bond length : Bond order (BO) is inversely proportional to the bond length (BL). i.e. higher the BO, smaller the BL

Information given by BO (contd.):

(vi) Magnetic properties:

1. If unpaired electron present in MO's, it is paramagnetic. Examples: O_2, O_2^+, O_2^-, NO^-, CN etc. Greater the number of unpaired electrons, the more will be its paramagnetic character. The elements having this property are known as ferromagnetic (strong attraction by magnetic field).

   Examples: Fe, Co, Ni etc.

2. If there is no unpaired electron in MO's, it will diamagnetic in nature. Examples: N_2, O_2^{2-}, NO^+, CO etc.

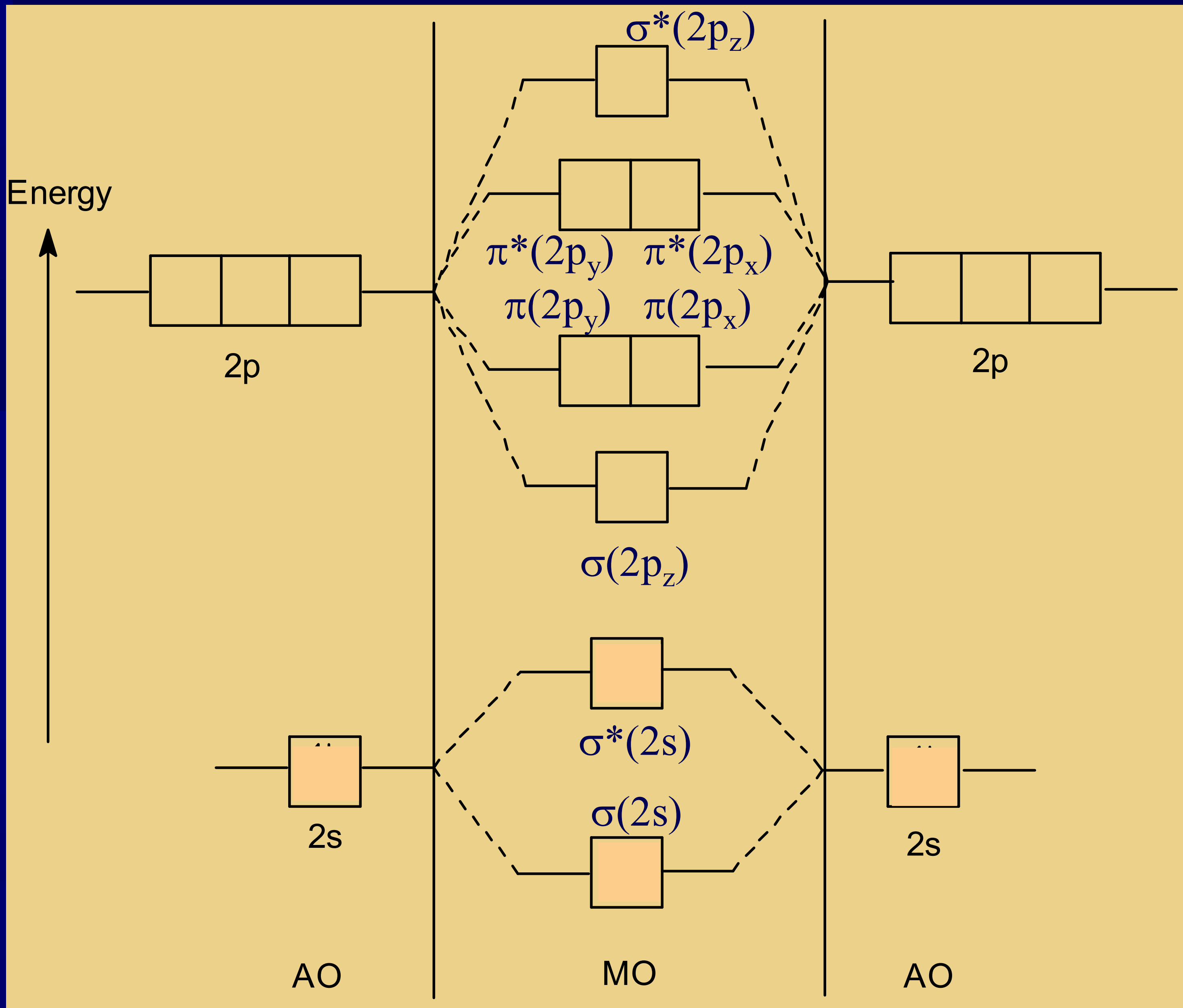
Molecular Orbital Diagram

- For Homonuclear Molecule such as- H_2, N_2, O_2, Ne_2 etc.
- For Heteronuclear Molecule such as- NO, CO, CN etc.

Objective:

1. Bond order
2. Magnetic properties
3. Electronic configuration

Basic Skeleton:

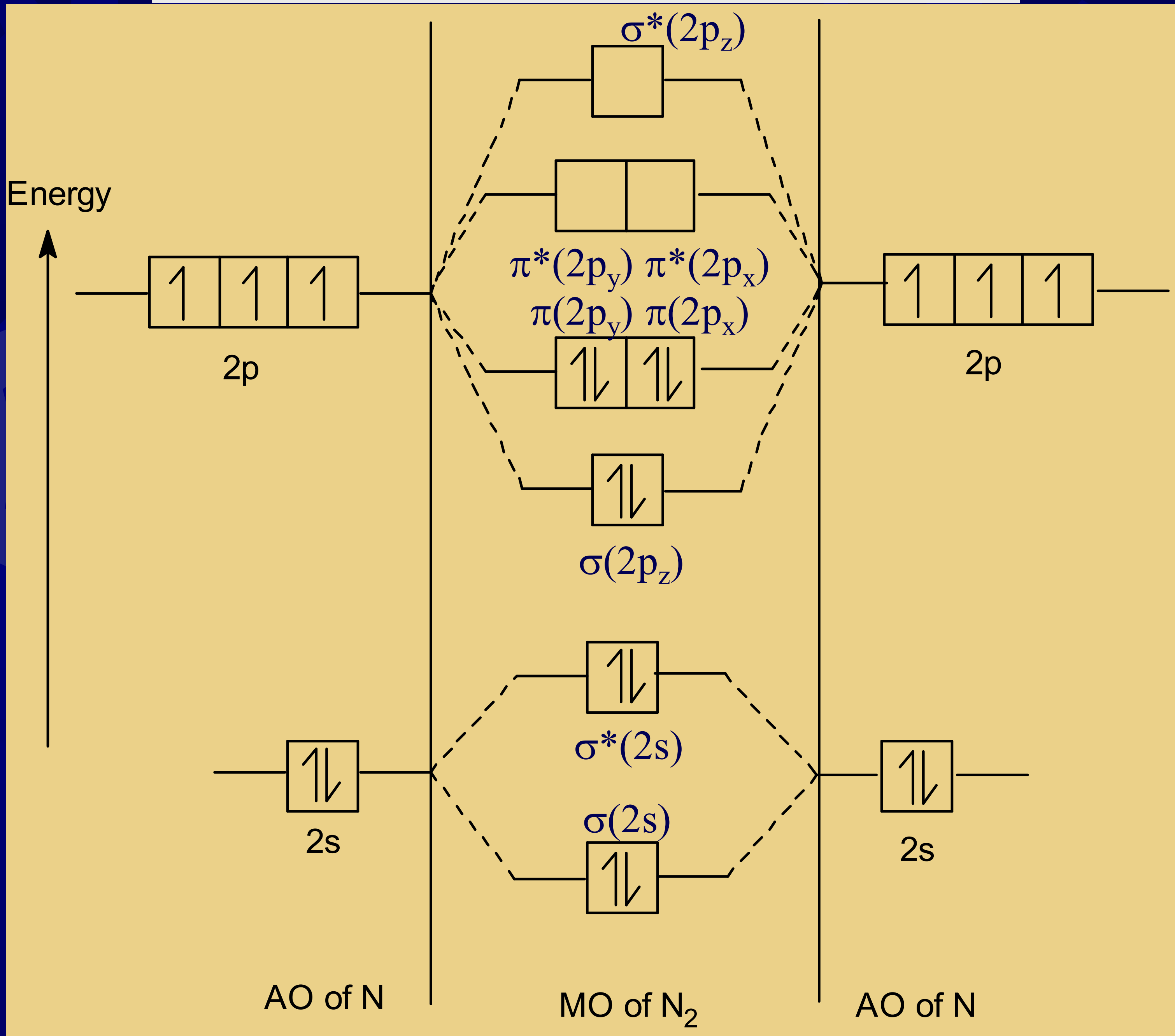


Molecular Orbital Diagram

Example- N_2

- Electronic configuration of N atom: $N(7)$- $1s^2\ 2s^2\ 2p_x^1 2p_y^1 2p_z^1$
- Number of valence electrons for one atom of N is 5.
- Total number of valence electrons for two atoms of N is 10.

Figure: Molecular orbital diagram of N_2 molecule

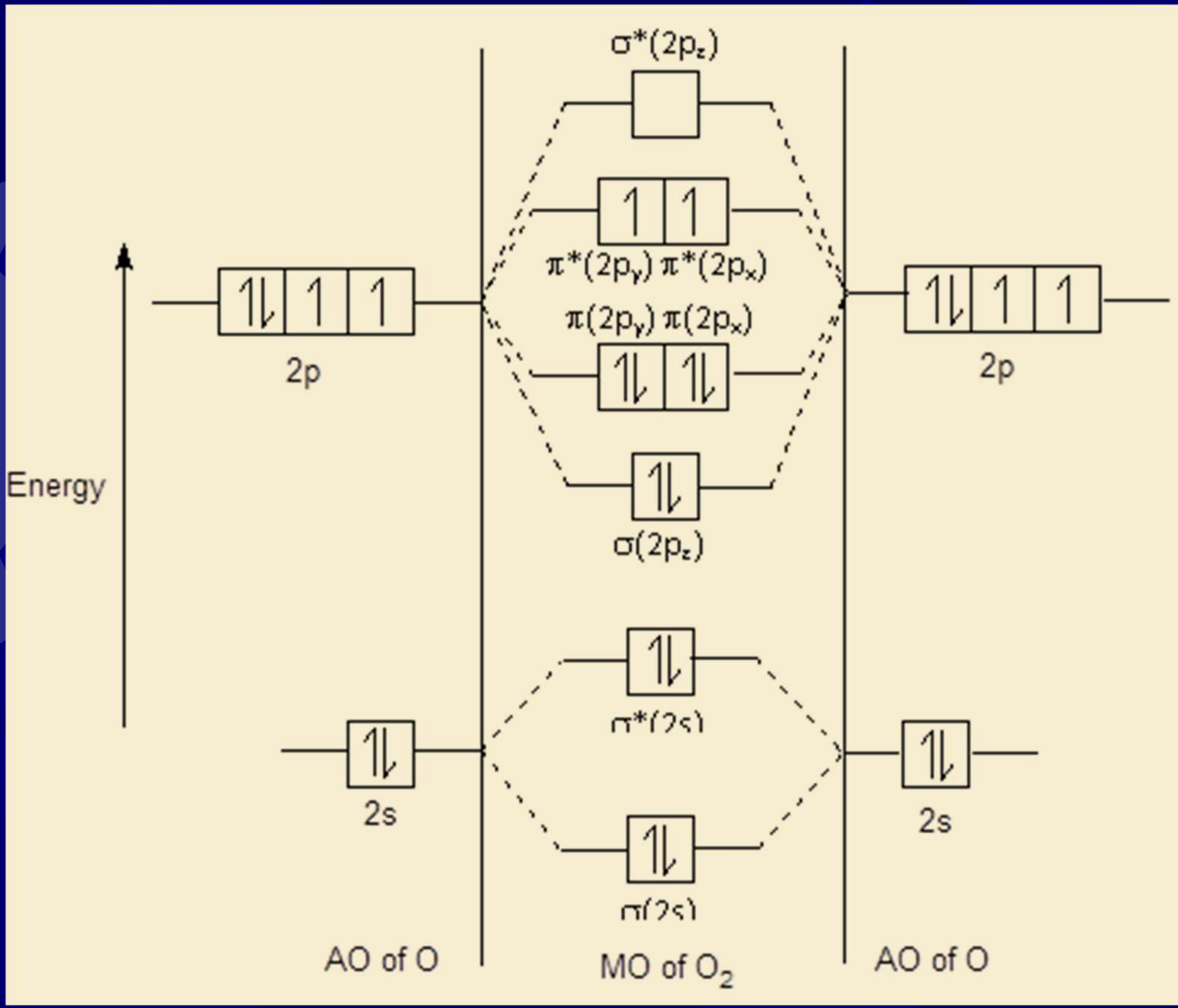


Information from MO diagram of N_2:

- Bond order is 3, three covalent bond exist and hence stable.
- Magnetic properties: diamagnetic; since there is no unpaired electron.
- Electronic configuration:

  $N_2(10)=KK\ \sigma(2s)^2\ \sigma^*(2s)^2\ \sigma(2p_z)^2\ \pi(2p_y)^2\ \pi(2p_x)^2$

Molecular Orbital Diagram for O_2:

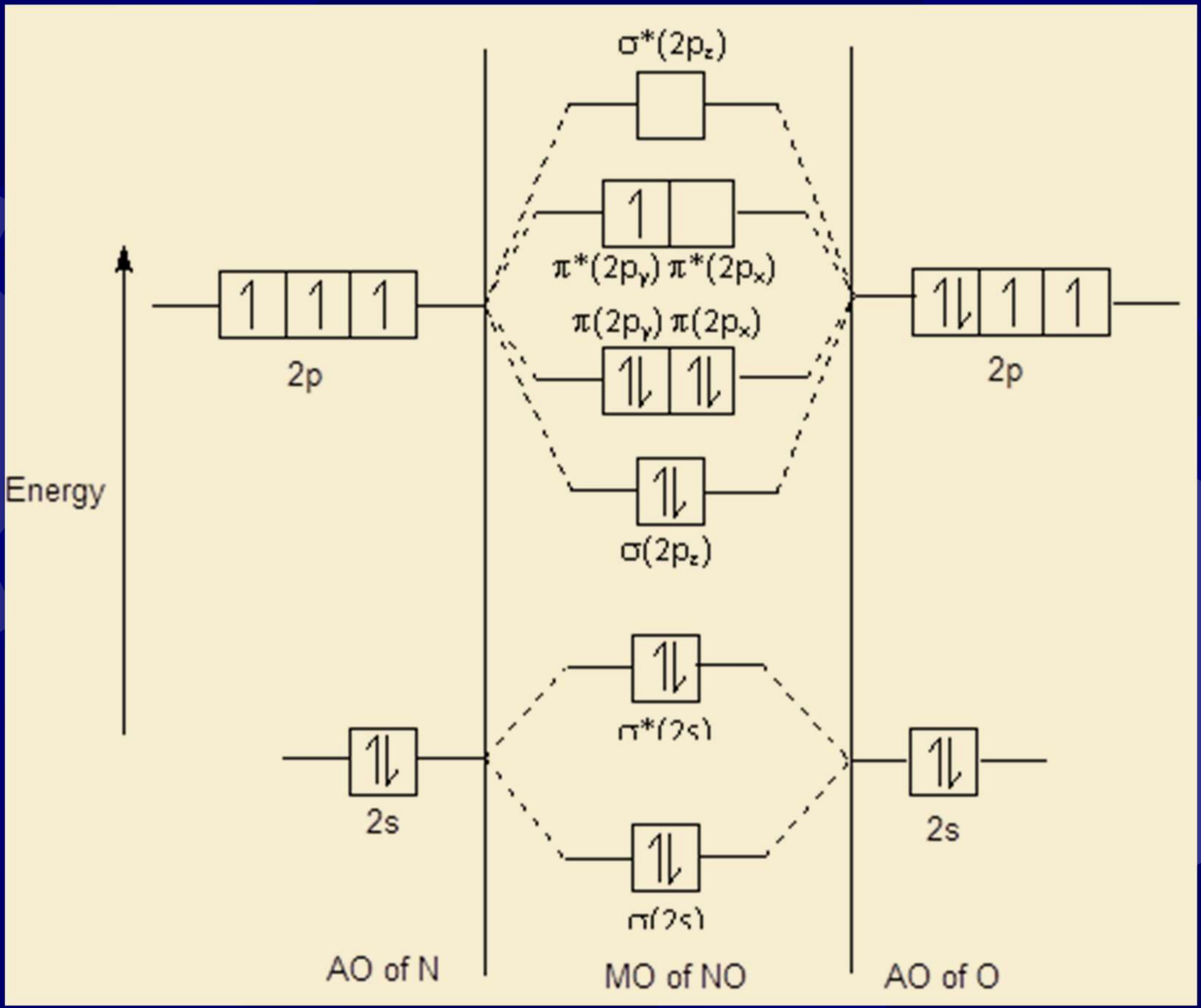


Information from MO diagram of O_2:

- Bond order is 2, two covalent bond exist and hence stable.
- Magnetic properties: paramagnetic; since there are two unpaired electrons in MO's.
- Electronic configuration:

$$O_2(12)=KK\ \sigma(2s)^2\ \sigma^*(2s)^2\ \sigma(2p_z)^2\ \pi(2p_y)^2\ \pi(2p_x)^2\ \pi^*(2p_y)^1\ \pi^*(2p_x)^1$$

Molecular Orbital Diagram for NO:

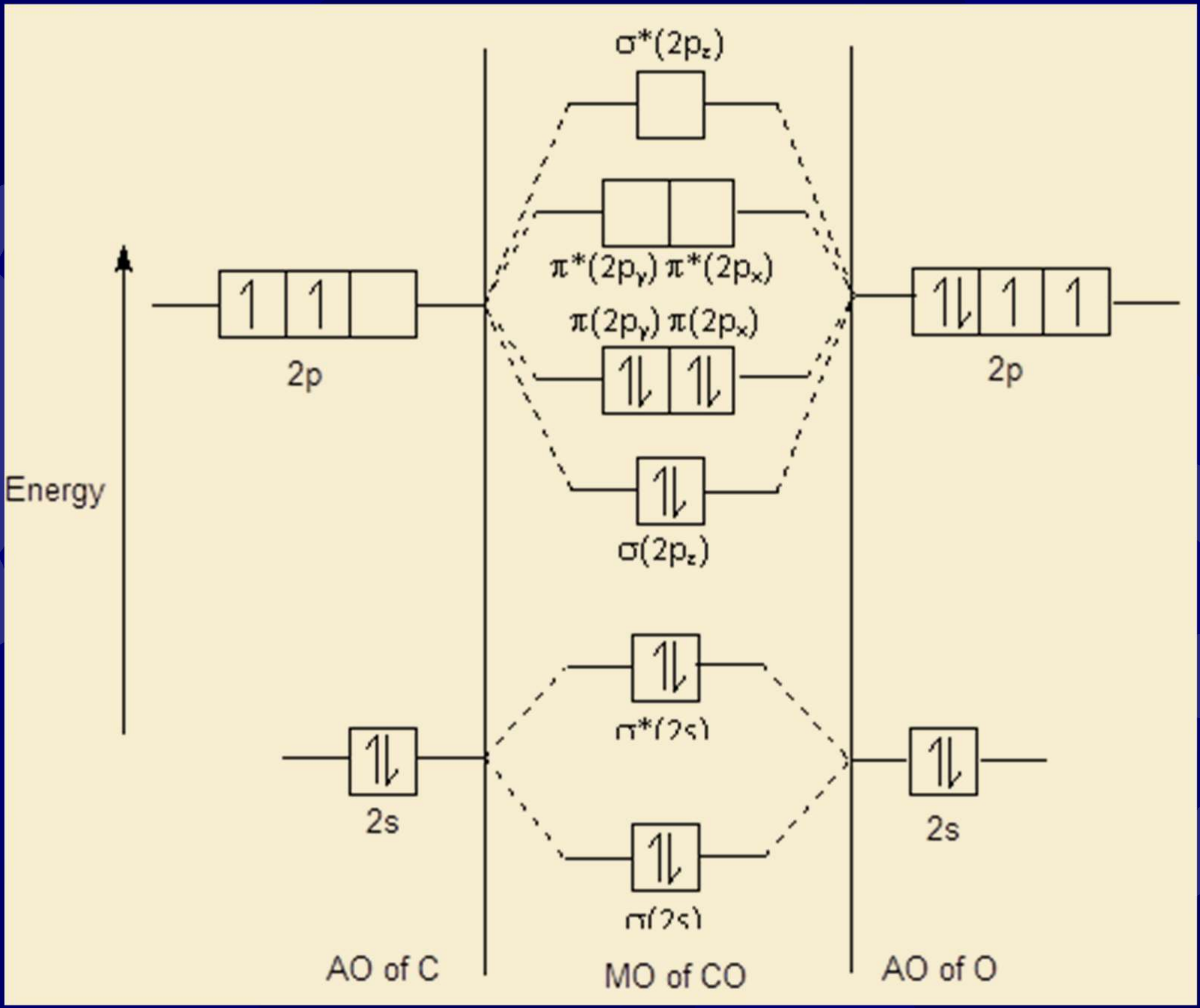


Information from MO diagram of NO:

- Bond order is 2.5, covalent bond exist and hence stable.
- Magnetic properties: paramagnetic; since there is one unpaired electrons in MO's.
- Electronic configuration:

NO (11)=KK $\sigma(2s)^2\ \sigma^*(2s)^2\ \sigma(2p_z)^2\ \pi(2p_y)^2\ \pi(2p_x)^2\ \pi^*(2p_y)^1$

Molecular Orbital Diagram for CO:



Information from MO diagram of CO:

- Bond order is 3, three covalent bond exist and hence stable.
- Magnetic properties: diamagnetic; since there is no unpaired electron.
- Electronic configuration:

CO (10)=KK $\sigma(2s)^2\ \sigma^*(2s)^2\ \sigma(2p_z)^2\ \pi(2p_y)^2\ \pi(2p_x)^2$